All four exceptional Seiffert resultant-mean inequalities

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Abstract

We prove the exceptional-parameter sign conjecture for the generalized Seiffert mean, including both signs of the outer power-mean parameter. A mixed derivative reduces all four cases to a one-variable inequality. Rigorous Taylor enclosures, together with analytic bounds at zero and infinity, prove that inequality on the entire positive axis.

Task 1: the original problem

The mean in the source is

$$S_\alpha(a,b) = \frac{\alpha(b-a)}{2\arctan\big(\tanh(\alpha\log(b/a)/2)\big)} \quad (a,b>0,\ a\neq b), \qquad S_\alpha(a,a)=a.$$

It is symmetric, homogeneous, and unchanged on replacing α by $-\alpha$. We use the source's formula: the database calls this a Stolarsky mean, but it is the generalized Seiffert family in [1, Eq. (1.4)]. Set

$$B_r(a,b) = \left(\frac{a^r+b^r}{2}\right)^{1/r}, \qquad \mathcal{R}(B_p,S_\alpha,B_q)(a,b) = B_p\big(S_\alpha(a,B_q(a,b)),S_\alpha(B_q(a,b),b)\big).$$

Theorem 1. *Let*

$$\alpha_\pm = \frac{1}{2}\sqrt{2\pm\sqrt{3}}.$$

For each row in the following table, and for all distinct positive a,b, one has

$$S_\alpha(a,b) < \mathcal{R}(B_p,S_\alpha,B_q)(a,b).$$

$\lvert\alpha\rvert$	p	q
α_-	$\sqrt{3}-1$	$2-\sqrt{3}$
α_-	$1-\sqrt{3}$	1
α_+	$1+\sqrt{3}$	1
α_+	$-1-\sqrt{3}$	$2+\sqrt{3}$

These are exactly all solutions of the source's conditions

$$q = \frac{1-p}{2}+2\alpha^2, \qquad p^2 = \frac{1-12\alpha^2+112\alpha^4-64\alpha^6}{1+4\alpha^2}, \qquad |\alpha|=\alpha_\pm.$$

Thus the theorem proves the entire original Conjecture 4.4. The published Conjecture 34, p. 21, retains only the two positive-p rows; those are included. The diagonal has equality as usual.

The mixed derivative

Assume $\alpha > 0$ and $a < b$. All four q's are positive. Put $H(a,b) = (b^q - a^q)S_\alpha(a,b)^p$. For $a < c < b$,

$$H(a,c) + H(c,b) - H(a,b) = \int_a^c \int_c^b H_{ab}(u,v)\, dv\, du. \tag{1}$$

It suffices to prove that H_{ab} has the strict sign of p. Indeed, take $c = B_q(a,b)$. The increments $c^q - a^q$ and $b^q - c^q$ are equal, so (1) says that

$$\frac{S_\alpha(a,c)^p + S_\alpha(c,b)^p}{2} - S_\alpha(a,b)^p$$

has the strict sign of p. Taking the p-th root gives Theorem 1 for either sign.

For $t > 0$, let

$$v(t) = \log \frac{\alpha \sinh(t/2)}{\arctan(\tanh(\alpha t/2))}, \qquad g(t) = \frac{\alpha}{2\cosh(\alpha t)\arctan(\tanh(\alpha t/2))}.$$

Then

$$v' = \tfrac{1}{2}\coth(t/2) - g, \tag{2}$$

$$v'' = -\frac{1}{4\sinh^2(t/2)} + \alpha \tanh(\alpha t) g + g^2. \tag{3}$$

Define

$$T(t) = v'' + p(v')^2 + q\coth(qt/2)v' - \frac{p+2q}{4}. \tag{4}$$

Write $s = p + q$, $t = \log(b/a)$, and

$$h(t) = e^{st/2} 2\sinh(qt/2) e^{pv(t)}.$$

Direct differentiation gives

$$H(a,b) = a^s h(t), \qquad h'' - sh' = phT, \qquad H_{ab}(a,b) = -\frac{a^s}{ab} ph(t)T(t).$$

Here $h(t) > 0$. It remains to show $T(t) < 0$ for every $t > 0$. We prove this for each of the four rows, with $F = -T$.

The neighborhood of zero

Put $A = \alpha^2$. Since

$$\mu(t) = \frac{2}{\alpha}\arctan(\tanh(\alpha t/2)), \qquad \mu'(t) = \operatorname{sech}(\alpha t),$$

we have $v' = \frac{1}{2}\coth(t/2) - \mu'/\mu$. Expansion at zero gives

$$\begin{aligned} v'(t) = {} & (1/12 + A/3)t - (1/720 + A^2/9)t^3 \\ & + (1/30240 + 38A^3/945)t^5 \\ & - (1/1209600 + 43A^4/2835)t^7 + O(t^9). \end{aligned}$$

Substitution in (4) gives, for $\alpha = \alpha_\sigma$,

$$F(t) = \frac{97 + \sigma 56\sqrt{3}}{201600} t^6 + O(t^8), \qquad \sigma \in \{-1, 1\}, \tag{5}$$

independently of the sign of p. The coefficients of degrees zero, two and four vanish exactly. Both leading coefficients are positive, since $97^2 - 3 \cdot 56^2 = 1$. The verifier checks these algebraic identities symbolically.

Here is an explicit analytic remainder bound. For complex z, set

$$\rho(z) = \frac{\mu(z)}{z}, \qquad U(z) = \frac{\operatorname{sech}(\alpha z)}{\rho(z)}, \qquad D(w) = w \coth w.$$

All three have removable value one at zero. The Taylor coefficients of $\sec z$ are nonnegative, and those of $\operatorname{sech} z$ have the same absolute values. This also follows directly from $(\tan z)' = 1 + \tan^2 z$ and $(\sec z)' = \sec z \tan z$. Since $0 < \alpha < 1$, for $|z| \le 1$ we obtain

$$|\rho(z) - 1| \le \int_0^1 (\sec u - 1)\, du = \log(\sec 1 + \tan 1) - 1 < \frac{1}{3}, \qquad |\operatorname{sech}(\alpha z) - 1| < 1.$$

It follows that $|U - 1| < 2$. Standard power-series estimates give

$$|D(z/2) - 1| < \frac{1}{4} \quad (|z| \le 1), \qquad |D(w)| < 2 \quad (|w| \le 1).$$

For completeness, these follow by bounding $\cosh w$ and $\sinh w/w - 1$ by their absolute Taylor sums, using

$$\cosh(1/2) - 1 < 1/7, \quad 2\sinh(1/2) - 1 < 1/20, \quad \cosh 1 < 8/5, \quad \sinh 1 - 1 < 1/5.$$

The same estimates exclude denominator zeros. As $v'(z) = (D(z/2) - U(z))/z$, parity and the maximum principle give

$$|v'(z)/z| < 9/4 \quad (|z| \le 1).$$

Cauchy's derivative estimate yields $|v'| < 9/8$ and $|v''| < 9/2$ on $|z| \le 1/2$. In all four cases $|p| < 3$, $0 < q < 4$, and $0 < p + 2q < 5$. Consequently F is even and analytic on that disc, and

$$|F(z)| < \frac{9}{2} + 3(9/8)^2 + 9 + \frac{5}{4} < 32.$$

If $F(t) = \sum c_k t^k$, then $|c_k| \le 32\, 2^k$. Using parity and (5), for $0 < t \le 1/4$ we have

$$\left| \frac{F(t) - \sum_{k=3}^{30} c_{2k} t^{2k}}{t^6} \right| \le \frac{32\, 2^{62} t^{56}}{1 - 2t} \le 2^{-44}. \tag{6}$$

The accompanying verifier computes the coefficients through degree 60 using rigorous formal ball series at 192-bit midpoint precision. For each of the four rows, Horner evaluation of $\sum_{k=3}^{30} c_{2k} y^{k-3}$ on $0 \le y \le 1/16$, minus 2^{-44}, is strictly greater than 10^{-8}. Thus $F(t) > 10^{-8} t^6 > 0$ throughout $0 < t \le 1/4$. No truncation is used without the analytic bound (6).

The compact interval

For $[l,u] \subset [1/4, 20]$, put $m = (l+u)/2$, $r = (u-l)/2$. Taylor's theorem gives, with $d = t - m$,

$$F(t) = \sum_{j=0}^{7} \frac{F^{(j)}(m)}{j!} d^j + \frac{F^{(8)}(\xi)}{8!} d^8, \qquad \xi \in [l,u]. \tag{7}$$

Evaluating (2)–(4) on the formal series $m + z$ encloses the first eight coefficients. Evaluating it on $I + z$, where the constant coefficient I encloses all of $[l,u]$, encloses $F^{(8)}(\xi)/8!$ for every possible ξ. Interval Horner evaluation of (7) on $d \in [-r, r]$ therefore covers every point of $[l,u]$.

The algorithm starts with the 79 intervals $[j/4, (j+1)/4]$, $1 \le j \le 79$, and bisects any interval whose enclosure is not strictly positive. Exact rational endpoints are used throughout. The computation terminates with:

α	p	accepted intervals
α_-	$1 - \sqrt{3}$	155
α_-	$\sqrt{3} - 1$	154
α_+	$-1 - \sqrt{3}$	554
α_+	$1 + \sqrt{3}$	139

Every leaf enclosure has lower endpoint greater than 10^{-12}. For each row, the leaves have disjoint interiors and their exact lengths sum to $79/4$. Thus $F > 0$ on all of $[1/4, 20]$. The certificate records derivative order, coefficient balls, partition hashes and counts. This is a whole-interval proof, not a finite sampling argument.

The infinite tail

Set $\epsilon = (e^t - 1)^{-1}$, $\delta = 2q/(e^{qt} - 1)$, $d = \alpha \tanh(\alpha t)$, and $s = p + q$. Equations (2)–(4) give the exact identity

$$\begin{aligned} T = {} & (d-s)g + (p+1)g^2 + (s-1)\epsilon + (p-1)\epsilon^2 \\ & - 2p\epsilon g + \delta(1/2 + \epsilon - g). \end{aligned} \tag{8}$$

For $t \ge 20$, the following uniform bounds hold:

$$\frac{5}{4}\alpha e^{-\alpha t} < g < \frac{4}{3}\alpha e^{-\alpha t}, \qquad \epsilon < 10^{-6}, \qquad \epsilon < \frac{101}{100} e^{-t}. \tag{9}$$

Indeed $\alpha t > 5$, $\arctan(\tanh(5/2)) > 3/4$, and

$$\frac{4}{\pi(1 + e^{-10})} > \frac{5}{4}.$$

These prove the bounds for g directly from its definition. Also $d < \alpha$. Useful rational parameter bounds are

$$\frac{129}{500} < \alpha_- < \frac{13}{50}, \quad \frac{1}{4} < 2 - \sqrt{3} < \frac{67}{250}, \quad 2 - \sqrt{3} - \alpha_- > \frac{9}{1000},$$
$$\frac{24}{25} < \alpha_+ < \frac{97}{100}, \qquad \frac{37}{10} < 2 + \sqrt{3} < 4.$$

The certificate verifies these fixed comparisons with strict ball enclosures. All subsequent bounds are uniform in t.

The positive-p, lower-α case. Here $s = 1$, $0 < p < 1$, and $q = 2 - \sqrt{3}$. Dropping negative terms in (8), and using $g < 1/250$, gives

$$T < -\frac{73}{100}g + \frac{203}{100}q(1/2 + \epsilon)e^{-qt}.$$

For the second term we used $qt > 5$ and $2/(1 - e^{-5}) < 203/100$. Thus

$$e^{\alpha t}T < -\frac{73}{100}\frac{5}{4}\frac{129}{500} + \frac{203}{100}\frac{501}{1000}\frac{67}{250}\frac{21}{25} < 0,$$

where $e^{-(q-\alpha)t} < e^{-9/50} < 21/25$. In the first term, $1 - \alpha - (p + 1)g > 73/100$.

The positive-p, upper-α case. Here $q = 1$, $s = 2 + \sqrt{3}$, and $\delta = 2\epsilon$. Combining the epsilon terms in (8) gives

$$T < -\frac{13}{5}g + \frac{401}{100}\epsilon.$$

Indeed $s - \alpha > 27/10$, $p + 1 < 4$, and $g < 1/75$; also $s\epsilon + (p + 1)\epsilon^2 < (401/100)\epsilon$. It follows that

$$e^{\alpha t}T < -\frac{78}{25} + \frac{401}{100}\frac{101}{100}\frac{3}{5} < 0,$$

using $1 - \alpha > 3/100$ and $e^{-3/5} < 3/5$.

The negative-p, lower-α case. Write $R = -p = \sqrt{3} - 1$. Here $q = 1$, $s = 2 - \sqrt{3}$, and $p + 1 = s$. From (8),

$$T < -\big(s - \alpha - sg - 2R\epsilon\big)g + 2\epsilon(1/2 + \epsilon) < -\frac{7}{1000}g + 3e^{-t}.$$

The coefficient estimate follows from $s - \alpha > 9/1000$, $s < 27/100$, $g < 1/250$, $R < 1$, and $\epsilon < 10^{-6}$. Since $1 - \alpha > 74/100$,

$$e^{\alpha t}T < -\frac{7}{1000}\frac{129}{400} + \frac{3}{10000} < 0.$$

Here $e^{-74/5} < 10^{-4}$.

The negative-p, upper-α case. Now $R = -p = 1 + \sqrt{3} < 3$, $q = 2 + \sqrt{3}$, and $s = 1$. The g^2 and ϵ^2 terms in (8) are negative. Hence

$$T < -\big(1 - \alpha - 2R\epsilon\big)g + \delta(1/2 + \epsilon) < -\frac{29}{1000}g + 16e^{-qt}.$$

Using $q - \alpha > 27/10$, we conclude

$$e^{\alpha t}T < -\frac{29}{1000}\frac{6}{5} + \frac{16}{10000} < 0,$$

since $e^{-54} < 10^{-4}$.

All exponential comparisons just used admit elementary rational certificates: $e^x > 1 + x + x^2/2$ gives the comparisons at $9/50$ and $3/5$; and $(74/5)^8/8! > 10000$, $54^4/4! > 10000$ give the other two. The four final rational differences are checked exactly by the verifier. This proves $T < 0$ for every $t \geq 20$. Together with the earlier intervals, $T < 0$ on all of $(0, \infty)$, completing the proof of Theorem 1. $\square$

Proof status. This is a computer-assisted proof. Its rigorous Taylor enclosures are substantive parts of the argument. The accompanying `verify_inequality.py` uses python-flint's Arb arithmetic [2] and SymPy for exact coefficient identities; it recreates `exact-certificate.json`. The authoring agent audited the proof and executed the verifier. Independent peer review and proof-assistant formalization are not claimed.

Proposed follow-up questions

These four questions are unproved directions, not additional solves.

Task 2. Find an analytic proof of the sign of T that eliminates the finite interval computation for all four parameter rows.

Task 3. Determine for which $0 < \alpha < 1$ the source's asymptotically optimal parameter pairs give a global resultant-mean comparison.

Task 4. Determine the sharp normalized gap for arbitrary split weights $0 < \lambda < 1$, replacing the equal q-power increments by $\lambda, 1-\lambda$, and find its asymptotics as a weight tends to zero. The qualitative comparison already follows from (1).

Task 5. Classify smooth homogeneous means M for which a constant sign of the mixed derivative of $(b^q - a^q)M(a,b)^p$ is necessary, as well as sufficient, for all weighted resultant comparisons.

References

[1] L. Mihoković and M. Raïssouli, *Results on comparison and sub/super-stabilizability of some new means*, arXiv:2405.13618v2 (2024), Conjecture 4.4; Applicable Analysis and Discrete Mathematics **20** (2026), 1–23, doi:10.2298/AADM240618001M, Conjecture 34. Complete published text: journal PDF.

[2] The python-flint developers, *Arb real numbers*, official documentation. The certificate records the installed software versions.